

$$S = 60 e^{2 \ln(\frac{2}{3}) t}$$

$$\ln \frac{1}{12} = \ln \left(\frac{2}{3}\right)^2 t$$

$$\frac{1}{12} = \frac{4}{9} t$$

4.)

$$\frac{9}{48} = t$$

ln

ln

Q3). $y = c_1 x + c_2 x \ln x \quad (0, \infty)$

$$x^2 y'' - x y' + y = 0$$

$$y(1) = 3, \quad y'(1) = -1$$

$$\ln \frac{2}{3} + \ln t$$

$$e^{1+t} e^{\ln \frac{2}{3}}$$

a.

$$3 = c_1 + c_2 \ln(1)$$

$$\boxed{c_1 = 3}$$

$$y' = c_1 + c_2 (\cancel{\ln x} + \cancel{\frac{1}{x}})$$

$$y' = c_1 + c_2 (\ln x + 1)$$

$$-1 = c_1 + c_2 \ln x + c_2$$

$$-1 = c_1 + c_2$$

$$-1 - 3 = c_2$$

$$\boxed{c_2 = -4}$$

$$1 = c_1$$

$$-1 = c_1 \cos \pi + c_2 \sin \pi$$

$$-1 = -c_1 e^{\pi}$$

$$-1 = -e^{\pi}$$

$$\hat{e} = 1$$

$$y = 3x - 4x \ln x$$

$$17) f_1(x) = 5, f_2(x) = \cos^2 x, f_3(x) = \sin^2 x$$

$$W(f_1, f_2, f_3) = \begin{vmatrix} 5 & \cos^2 x & \sin^2 x \\ 0 & -\sin^2 x & \cos^2 x \\ 0 & -\cos^2 x & -\sin^2 x \end{vmatrix}$$

$$5 \begin{vmatrix} -\sin^2 x & \cos^2 x \\ -\cos^2 x & -\sin^2 x \end{vmatrix}$$

$$5(\sin^4 x + \cos^4 x)$$

$$\textcircled{1+1} \quad 5(1) = 5$$

$$W(f_1, f_2, f_3) = \begin{vmatrix} 5 & \cos^2 x & \sin^2 x \\ 0 & 2(\cos x)(-\sin x) & 2\sin x \cos x \\ 0 & 2\sin^2 x - 2\cos^2 x & 2\cos^2 x + 2\sin^2 x \end{vmatrix}$$

$$5 \begin{vmatrix} -2\cos x \sin x & 2\sin x \cos x \\ 2\sin^2 x - 2\cos^2 x & 2\cos^2 x + 2\sin^2 x \end{vmatrix}$$

$$5 \left[(-2\cos x \sin x)(2\cos^2 x + 2\sin^2 x) - (2\sin x \cos x)(2\sin^2 x - 2\cos^2 x) \right]$$

$$5(-4\cos^3 x \sin x + 4\cos x \sin^3 x)$$

$$- (4\sin^3 x \cos x - 4\sin x \cos^3 x)$$

$$5(-4\cos^3 x \sin x + 4\sin^3 x \cos x - 4\sin^3 x \cos x + 4\cos^3 x \sin x)$$

$= 0$ dependent

$$19) f_1(x) = x, f_2(x) = x-1, f_3(x) = x+3$$

$$W(f_1, f_2, f_3) = \begin{vmatrix} x & x-1 & x+3 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{vmatrix}$$

$$x \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix} - 1 \begin{vmatrix} x-1 & x+3 \\ 0 & 0 \end{vmatrix}$$

$$0 - 0 = 0 \quad \text{dependent}$$

$$20) f_1(x) = 2+x, f_2(x) = 2+|x| = 2+x, f_3(x) = 2-x$$

$$W(f_1, f_2) = \begin{vmatrix} \cancel{2+x} & \cancel{2+x} \\ 2+x & 2+x & 2-x \\ 1 & 1 & -1 \\ 0 & 0 & 0 \end{vmatrix}$$

$$2+x \begin{vmatrix} 1 & -1 \\ 0 & 0 \end{vmatrix} - 1 \begin{vmatrix} 2+x & 2-x \\ 0 & 0 \end{vmatrix}$$

$$0 - 0 = 0 \quad \text{dependent}$$

$$25) y'' - 2y' + 5y = 0$$

$$y_1 = e^x \cos 2x, y_2 = e^x \sin 2x \quad (-\infty, \infty)$$

$$y' = e^x (-\sin 2x)(2) + \cos 2x e^x$$

$$y'_0 = -2e^x \sin 2x + e^x \cos 2x$$

$$= e^x (\cos 2x - 2\sin 2x)$$

$$y' = e^x [-\sin 2x(2) - 2(\cos 2x)(2)] + \cos 2x e^x + 2\sin 2x e^x$$

$$y'' = -2e^x \sin 2x - 4e^x \cos 2x + 2\cos 2x - 2\sin 2x e^x$$

$$y'' = -4e^x \sin 2x - 3e^x \cos 2x$$

$$-4e^x \sin 2x - 3e^x \cos 2x - 2(e^x \cos 2x - 2e^x \sin 2x) + 5(e^x \cos 2x)$$

$$-4e^x \sin 2x - 3e^x \cos 2x - 2e^x \cos 2x + 4e^x \sin 2x + 5e^x \cos 2x$$

$$0 = 0$$

$$y_2 = e^x \sin 2x$$

$$y'_1 = e^x 2 \cos 2x + \sin 2x e^x$$

$$y'_1 = 2e^x \cos 2x + e^x \sin 2x$$

$$y'_1 = e^x (2 \cos 2x + \sin 2x)$$

$$y''_1 = e^x [-4 \sin 2x + 2 \cos 2x] + (2 \cos 2x + \sin 2x) e^x$$

$$y''_1 = -4e^x \sin 2x + 2e^x \cos 2x + 2e^x \cos 2x + e^x \sin 2x$$

$$y''_1 = 4e^x \cos 2x - 3e^x \sin 2x$$

$$4e^x \cos 2x - 3e^x \sin 2x - 2(2e^x \cos 2x + e^x \sin 2x) + 5(e^x \sin 2x)$$

$$4e^x \cos 2x - 3e^x \sin 2x - 4e^x \cos 2x - 2e^x \sin 2x + 5e^x \sin 2x$$

$0 = 0$ so y_1, y_2 are the solutions of DE

$$W(y_1, y_2) = \begin{vmatrix} e^x \cos 2x & e^x \sin 2x \\ e^x (2 \cos 2x + \sin 2x) & e^x (2 \cos 2x + \sin 2x) \end{vmatrix}$$

$$e^x \cos 2x \left(e^x (2 \cos 2x + \sin 2x) \right) - e^x \sin 2x (e^x) \quad \cos 2x - 2 \sin 2x$$

$$2e^{2x} \cos^2 2x + e^{2x} \cos 2x \sin 2x - e^{2x} \cos 2x \sin 2x + 2e^{2x} \sin^2 2x$$

$$2e^{2x} (\cos^2 2x + \sin^2 2x)$$

so solutions are L. independent

$$\cancel{x dy + (y - 2x) dx = 0}$$

$$x dx + (y - 2x) dy = 0$$